

Assignment Title: Model Evaluation

Project Title: Fake News Detection

1. Reading Summary:

This week, I studied the topic of **Model Evaluation**, which focuses on measuring how well a trained machine learning model performs on unseen data.

- Learned about **evaluation metrics** such as **Precision, Recall, F1-score, and Accuracy**.
- Understood the use of **Confusion Matrix** to summarize model predictions.
- Explored **ROC (Receiver Operating Characteristic) curve** and **AUC (Area Under Curve)** to assess model performance visually.
- Understood why different evaluation metrics are important for different types of problems — especially for **imbalanced datasets** like Fake News Detection.

2. Task Performed:

- Loaded the trained models (Logistic Regression and Random Forest) from previous week.
- Used the **test dataset** to generate predictions.
- Created and visualized a **Confusion Matrix** using Seaborn's heatmap.
- Calculated the following evaluation metrics:
 - **Precision**
 - **Recall**
 - **F1-Score**
 - **Accuracy**
- Plotted the **ROC Curve** and calculated **AUC score** for both models.
- Compared performance between Logistic Regression and Random Forest.
- Wrote a short reflection explaining which metric is most important for this project.

3. Calculations and Analysis:

- **Confusion Matrix** showed how many fake and real news articles were correctly and incorrectly classified.
- **Precision** measured how many predicted fake news were actually fake.
- **Recall** indicated how well the model identified fake news from all actual fake articles.
- **F1-Score** balanced both Precision and Recall.
- **ROC-AUC Curve** displayed overall performance — closer to 1 means better classification.

Example Results (Approximate):

Metric	Logistic Regression	Random Forest
Accuracy	0.988	0.998
Precision	0.99	1.00
Recall	0.98	0.99
F1-Score	0.985	0.995
AUC Score	0.99	1.00

4. Learning Outcome:

From this task, I learned :

- How to evaluate classification models using different performance metrics.
- The importance of using **Precision, Recall, and F1-score** for imbalanced datasets.
- How to plot and interpret **ROC and AUC curves**.
- Why Accuracy alone can be misleading when false positives and false negatives matter.
- How to decide which evaluation metric best fits a project's goals.

5. Challenges Faced:

- Confusion Matrix and ROC plots were difficult to interpret at first for imbalanced data.
- Calculating and comparing multiple metrics for two models required careful organization.

- ROC-AUC calculation needed proper probability outputs (predict_proba) rather than direct class predictions.
- Deciding which metric (Precision, Recall, F1) is most important for the project required understanding real-world implications of misclassifying fake news.

6. Reflection: Which Metric Is Most Important and Why?

For my **Fake News Detection** project, **Recall** is the most important metric:

- It measures how well the model identifies **all fake news**.
- Missing fake news (false negatives) can spread misinformation, so **higher recall** is critical.
- F1-score is also important to balance precision and recall for overall performance.

7. Results Summary:

Model	Accuracy	Precision	Recall	F1-Score	Key Insight
Logistic Regression	0.988	0.99 / 0.98	0.98 / 0.99	0.985	Strong baseline; interpretable; slightly lower recall than RF
Random Forest	0.998	1.00 / 1.00	0.99 / 1.00	0.995	Nearly perfect classification; captures complex patterns

8. GitHub Repository Link:

GitHub repository link here:

<https://github.com/AlmasMalik66/DataScience-AI-Assignments/tree/main/Week7>